Résumé

Md Shajalal, PhD

Postdoctoral Research Fellow, IT Security and Consumer Computing
Faculty III, University of Siegen

Siegen, Germany
Personal Webpage
md.shajalal@uni-siegen.de

Research Interests

Explainable AI (XAI) and Human-Centered AI, Applied Machine/Deep Learning, Multimodal Signal Anal- ysis (text, audio, image), Natural Language Processing (NLP) and Information Retrieval (IR) — with a growing research direction toward explainability and human-centered evaluation for multimodal and vision- language systems.

Education

Nov 2021 – Mar 2025 | **PhD in Explainable AI (XAI)**
University of Siegen, Germany
Thesis: *"Towards Human-centered Actionable Explainable AI-enabled Systems"*
Grade: **Summa Cum Laude** (**PhD with Highest Distinction**)
Award: IHK **Best PhD Dissertation Award** by University of Siegen

March 2023 – May 2023 | **Visiting PhD Scholar**
University of Strathclyde, Glasgow, UK
Project: *"Explainable Energy Recommender"*

Sep 2017 | **Master of Engineering** in Computer Science and Engineering
Toyohashi University of Technology, Toyohashi, Aichi, Japan.
Thesis: *"Mining and Diversifying Query Subtopics From Web Documents Focusing on Semantic Features* [**PDF**]"
Result: Earned **A** (**80-100 percent**) in all major courses

2015 | **B.Sc. (Engineering)** in Computer Science and Engineering
University of Chittagong, Chittagong-4331, Bangladesh.
CGPA: **3.84** /4.00, **Rank**: **2**nd among a graduating class of 44 Students.

Experience

Jan 2025– Present | **Postdoctoral Research Fellow**
IT-Security and Consumer Computing, Business Information and New Media
University of Siegen, Germany
Responsibilities: Conducting SOTA research in AI and XAI, Writing research proposals, Supervising Thesis and Teaching Bachelor and Masters Courses

Nov 2021 – Dec 2024 | **Marie Curie Research Fellow**
Fraunhofer Institute for Applied Information Technology, Germany &
MSCA GECKO ITN funded by EU

Dec 2022 – Present | **Visiting Researcher**
AGENTS.Inc, Berlin, Germany

Nov 2021 – Mar 2025 | **PhD Researcher**
IT-Security and Consumer Computing
Information Systems and New Media
University of Siegen, Siegen, Germany

Jul 2019– Oct 2021 | **Faculty Member in Computer Science**
Department of Computer Science and Engineering
Hajee Mohammad Danesh Sci. & Tech. University, Bangladesh
Responsibilities: Conducting SOTA research in AI and XAI, Supervising Thesis and Teaching Bachelor and Masters Courses

Oct 2028 – Jun 2019 | **Lecturer of Computer Science**
Department of Computer Science and Engineering
Bangladesh Agricultural University, Mymensingh, Bangladesh

Nov 2017 – Sep 2018 | **Lecturer of Computer Science**
Department of Computer Science and Engineering
Bangladesh Army University of Science and Technology, Bangladesh

Oct 2015 – Sep 2017 | **Teaching and Research Assistant**
Knowledge Data Engineering and Information Retrieval Lab
Toyohashi University of Technology, Japan

May 2015 – Sep 2015 | **Software Engineer**
Kona Software Lab Limited, KONAI, South Korea
Z Tower, House #4, Road #132, Gulshan, Dhaka, Bangladesh

Selected Publication (Full List)

International Journal

[11] **Md Shajalal**, Alexander Boden and Gunnar Stevens. 2024. *ForecastExplainer*: Explainable Household Energy Demand Forecasting by Approximating Shapley Values Using DeepLIFT. *Technological Forecasting and Social Changes*, Elsevier, 206, pp. 123588 (**IF: 12.9, Q1**) [**PDF**]

[10] Chiara Tellarini, **Md Shajalal**, Nico Castelli, Martin Stein, Alexander Boden, Toke Haunstrup Christensen, "A mixed-method approach to study the impacts of energy micro-generation combined with appliance-level feedback on everyday practices", Energy Efficiency, Springer (**IF 4.00**) [**PDF**]

[9] **Md Shajalal**, Milad Bohlouli, Hari Prasanna Das, Alexander Boden and Gunnar Stevens. 2024. Improved Thermal Comfort Model Leveraging Conditional Tabular GAN Focusing on Feature Selection. *IEEE Access*, IEEE [IF: **3.9, Q1**] [**PDF**]

[8] **Md Shajalal**, Alexander Boden and Gunnar Stevens. 2022. Explainable Product Backorder Prediction Exploiting CNN: Introducing Explainable Models in Businesses. *Electronic Markets, Springer-Nature*. [**IF: 8.5, Q1**] [**PDF**]

[7] Md Rezaul Karim, Tanhim Islam, **Md Shajalal**, Oya Beyan, Christoph Lange, Michael Cochez, Dietrich Rebholz-Schuhmann and Stefan Decker. 2023. Explainable AI for Bioinformatics: Methods, Tools, and Applications. *Briefing in Bioinformatics, Oxford University Press*. [**IF:13.994, Q1**] [**PDF**]

[6] **Md Shajalal** and Masaki Aono. 2020. Coverage-based Query Subtopic Diversification Leveraging Semantic Relevance. *Knowledge and Information Systems (KAIS) 62, pp 2873–2891 (2020) Springer-Verlag London Ltd., part of Springer Nature*, April 2020. [**IF: 2.93 Q1**]

[**PDF**]

[5] **Md Shajalal** and Masaki Aono. 2019. Semantic Textual Similarity between Sentences Using Bilingual Word Semantics. *Progress in Artificial Intelligence, Springer-Verlag GmbH Germany, part of Springer Nature*, March 2019 [**Q1, IF: 4.2**] [**PDF**]

[4] **Md Shajalal**, Petr Hajek and Mohamad Zoynul Abedin. 2021. Product Backorder Prediction with Deep Neural Networks on Imbalanced Data. *International Journal of production research*, Taylor and Francis Volume-8, Issue-2, pp(263-272), Jun 2019 [**IF: 9.018, Q1**] [**PDF**]

[3] M. Atabuzzaman, **Md Shajalal**, M. E. Ahmed, M. I. Afjal and M. Aono. 2021. Leveraging grammatical roles for measuring semantic similarity between texts. *IEEE Access*, vol. 9, pp. 62972-62983, 2021, [**IF: 3.476**] [**PDF**]

[2] Machbah Uddin, Mhohammad Aminul Islam, **Md Shajalal**, Afzal Hossain and Sayeed Iftekhar, "Paddy Seed Variety Identification Using T20-HOG and Haralick Textural Features", *Complex and Intelligent Systems*, volume 8, pp. 657–671 (2022), Springer-Nature [**IF: 6.700**] [**PDF**].

[1] Mohammad Aminul Islam, Machbah Uddin, Md Rakib Hasan and **Md Shajalal**, "Germinative Paddy Seed Identification Using Deep Convolutional Neural network", Multimedia Tools and Applications, Springer-Nature, [**IF: 3.6**] [**PDF**].

International Conference (Peer-Reviewed)

[C32] **Md Shajalal**, Md Mahedy Riday, Sima Amirkhani, Gunnar Stevens"Human-Centered Explanations for Audio Deepfakes: Making Machine Reasoning Human-Perceptible Through Voice Traits", *HCI International 2026* (**HCII2026**), Montreal, Canada, Lecture Notes in Computer Science, Springer – **Best Paper Award (AI in HCI)**

[C31] Sidra Naveed, **Md Shajalal**, Gunnar Stevens, "From Algorithmic Transparency to Human Sensemaking: User Perceptions of Word-highlighting-Based Explanations for Fake Review Detection", *HCI International 2026* (**HCII2026**), Montreal, Canada, Lecture Notes in Computer Science, Springer.

[C30] Maksuda Bilkis Baby, Md Atabuzzman, **Md Shajalal**, "Attention-based Code Summarization for Multi-label Vulnerability Detection in Source Code", *24th International Conference on Cryptology and Network Security* (**CANS2025**), Osaka, Japan, Lecture Notes in Computer Science, Springer.

[C29] **Md Shajalal**, Md Atabuzzaman, Alexander Boden, Gunnar Stevens, Delong Du, "Do the Explanations Makr Sense? Explainable Fake Review Identification and Users' Perspectives on Explanations", *The 3rd International Conference on eXplainable Artificial Intelligence* (**xAI2025**), Istanbul, Turkey [**Accepted**]

[C28] **Jacob Larock, Md Shajalal**, Gunnar Stevens, "Interpretable Fake Voice Identification: A Hybrid Deep-Learning Model and Explanation Evaluation", *The 3rd International Conference on eXplainable Artificial Intelligence* (**xAI2025**), Istanbul, Turkey [**Accepted**]

[C27] **Shamima Rayhana, Md Shajalal**, Md Atabuzzaan, Gunnar Stevens, "Interpretable Sexism Detection with Explainable Transformers", *The 3rd International Conference on eXplainable Artificial Intelligence* (**xAI2025**), Istanbul, Turkey [**Accepted**]

[C26] **Md Shajalal**, Alexander Boden, Gunnar Stevens, Delong Du, Dean-Robin Kern, "Explaining AI Decisions: Towards Achieving Human-Centered Explainability in Smart Home Environments", The 2nd International Conference on eXplainable Artificial Intelligence (xAI2024), Malta, Springer LNCS [**PDF**]

[C25] Md. Atabuzzaman, Maksuda Bilkis Baby, and **Md Shajalal** "Transformer-based Bengali Textual Emotion Recognition", *Proceedings of the 20th International Conference on Natural Language Processing (ICON)*, India, ACM [**PDF**]

[C24] Md. Atabuzzaman, **Md Shajalal**, Maksuda Bilkis Baby, and Alexander Boden, "Arabic Sentiment Analysis with Noisy Deep Explainable Model", *7th International Conference on Natural Language Processing and Information Retrieval (ACM NLPIR2023)*, Seul, South Korea, ACM. [**Preprint**] (Equal Contribution)

[C23] Md. Rezaul Karim, **Md Shajalal**, Alex Graß, Till Döhmen, Sisay Adugna Chala, Christian Beecks and Stefan Decker, "Interpreting Black-box Models for High Dimensional Datasets", *The 10th IEEE International Conference on Data Science and Advanced Analytics (***IEEE DSAA 2023***)*, Thessaloniki, Greece **IEEE**, [**Core Ranked: A] (Accepted)**

[C22] Md. Rezaul Karim, Lina Comet, **Md Shajalal**, P. de Perthuis, D. Rebholz-Schuhmann, Stefan Decker, "From Large Language Models (LLMs) to Knowledge Graphs for Biomarker Discovery in Cancer", *The 57th Hawaii International Conference on System Sciences (HICSS 2023)*, Hawai, USA. [**Core Rank: A, Accepted**] [**Preprint**]

[C21] **Md Shajalal**, Sebastian Denef, Md. Rezaul Karim, Alexander Boden and Gunnar Stevens, "Unveiling the Black Box: Explainable Deep Learning Models for Patent Classification", *The 1st World Conference on eXplainable Artificial Intelligence (***xAI 2023***)*, Lisbon, Portugal, **Springer LNCS**, [**Accepted**]

[C20] **Md Shajalal**, Md. Atabuzzaman, Maksuda Bilkis Baby , Md Rezaul Karim and Alexander Boden, "Textual Entailment Recognition with Semantic Features from Empirical Text Representation", *International Conference on Speech and Language Technologies 2022* (SPELLL 2022) India, **Springer LNCS** [**PDF**].

[C19] Md Rezaul Karim, Suman Kanty Dey, Tanhim Islam, **Md Shajalal**, and BR Chakravarthi, "Multimodal hate speech detection from bengali memes and texts", *International Conference on Speech and Language Technologies 2022* (SPELLL 2022) India, **Springer LNCS** [**PDF**]

[C18] **Md Shajalal**, Milad Bohlouli, Hari Prasanna Das, Alexander Boden, and Gunnar Stevens, "Focus on What Matters: Improved Feature Selection Techniques for Personal Thermal Comfort Modelling", *The 9th ACM International Conference on Systems for Energy-Efficient Built Environments* (**ACM BuildSys 2022**), Boston, Massachusetts, USA, **ACM** [**PDF**].

[C17] **Md Shajalal**, Alexander Boden, Gunnar Stevens "Towards User-centered Explainable Energy Demand Forecasting System", *Thirteenth ACM International Conference on Future Energy Systems 2022* (ACM e-Energy'22), Massachusetts, USA, **ACM** [**PDF**]

[C16] Dean-Robin Kern, Gunnar Stevens, Erik Dethier, Sidra Naveed, Fatemeh Alizadeh, Du Delong, **Md Shajalal**, "Peeking Inside the Schufa Blackbox: Explaining the German Housing Scoring Syste", ACM CHI 2023 Workshop on Human-Centered Explainable AI (HCXAI), collocated with *ACM CHI Conference on Human Factors in Computing Systems 2023*, Hamburg, Germany. [**PDF**]

[C15] Maksuda Bilkis Baby, Bushra Ankhari, **Md Shajalal**, Md Atabuzzaman, Fazle Rabbi, Masud Ibn Afjal, "Identifying Duplicate Questions Leveraging Recurrent Neural Network", 4th International Conference on Trends in Computational and Cognitive Engineering (TCCE-2022), Bangladesh [**PDF**]

[C14] **Md Shajalal** and Masaki Aono "Semantic Sentence Modeling for Learning Textual Similarity Exploiting LSTM" *Proceedings of 2nd International conference on cyber security and computer science 2020 (ICONCS2020)*, **Springer Nature Lecture Notes LNICST**, *Springer-Verleg GmbH Germany*, [**PDF**]

[C13] **Md Shajalal**, Masaki Aono, and Muhammad Anwarul Azim, "Aspect-based Query Expansion for Search Results Diversification", *7th IEEE International Conference on Informatics, Electronics and Vision and 2nd Internation Conference on Imaging, Vision and Pattern Recognition (***IEEE ICIEVE-IcIVPR 2018***)**, Kitakyushu, Fukuaka, Japan, Jun 2018. [**PDF**]

[C12] **Md Shajalal** and Masaki Aono "Sentence-Level Semantic Textual Similarity using Word-Level Semantics" *IEEE International Conference on Computer and Electrical Engineering 2018 (***IEEE ICECE 2018***)**, BUET, Dhaka, Bangladesh. [**PDF**]

[C11] **Md Shajalal** and Masaki Aono "Semantic Textual Similarity in Bengali Text," *IEEE International Conference on Bangla Speech and Language Processing (***IEEE ICBSLP 2018***)**, SUST, Sylhet, Bangladesh, Sep. 2018. [**PDF**]

[C10] Md Zia Ullah, **Md Shajalal**, Abu Nowshed Chy, and Masaki Aono, "Query Subtopic Mining Exploiting Word Embedding for Search Result Diversification," *Twelfth Asia Information Retrieval Societies Conference (AIRS 2016)*, pp(308-314), Beijing, China, Nov. 2016, **Springer LNCS**.[**PDF**] (**Best Presentation Award**)

[C9] **Md Shajalal**, Md Zia Ullah, Abu Nowshed Chy, and Masaki Aono "Query Subtopic Diversification based on Cluster Ranking and Semantic Features," *IEEE International Conference on Advanced Informatics: Concepts, Theory and Application (***IEEE ICAICTA 2016***)**, pp(1-6), Penang, Malaysia, Aug. 2016. [**PDF**]

[C8] Md Elius Ahmed, **Md Shajalal** Md. Atabuzzaman and Masaki Aono, "Semantic similarity based on word recurrence ratio focusing on WordNet," *23rd IEEE International Conference on Computer and Information Technology (***IEEE ICCIT2020***)*, AUST, Dhaka, Bangladesh, 2020. [**PDF**]

[C7] Md. Atabuzzaman and **Md Shajalal**, "Measuring semantic similarity of Bengali texts with Parts-of-Speech tags and word-level semantics," *23rd IEEE International Conference on Computer and Information Technology (***IEEE ICCIT2020***)*, AUST, Dhaka, Bangladesh, 2020. [**PDF**]

[C6] Md Zia Ullah, **Md Shajalal**, and Masaki Aono, "KDEIM at NTCIR-12 IMine-2 Search Intent Mining Task: Query Understanding Through Diversified Ranking of Subtopics," *Proceedings of the Twelfth NTCIR Conference on Evaluation of Information Access Technologies,* NII, Tokyo, Japan, 2016. [**PDF**]

[C5] Abu Nowshed Chy, Md Zia Ullah, **Md Shajalal**, and Masaki Aono, "KDETM at NTCIR-12 Temporalia Task: Combining a Rule-based Classifier with Weakly Supervised Learning for Temporal Intent Disambiguation," *Proceedings of the Twelfth NTCIR Conference on Evaluation of Information Access Technologies,* Japan, 2016. [**PDF**]

[C4] **Md Shajalal** and Masaki Aono "Semantic Textual Similarity between Sentences Using Bilingual Word Semantics," *Progress in Artificial Intelligence, Springer-Verlag GmbH Germany, part of Springer Nature*, March 2019 [**Q1, IF: 4.2**] [**PDF**]

[C3] **Md Shajalal**, Petr Hajek and Mohamad Zoynul Abedin, "Product Backorder Prediction with Deep Neural Networks on Imbalanced Data" *International Journal of production research*, Taylor and Francis Volume-8, Issue-2, pp(263-272), Jun 2019 [**IF: 9.018, Q1**] [**PDF**]

[C2] **Md Shajalal** Mohammad Zoynul Abedin, Mohammed Mohi Uddin "Handling Class Imbalanced Data in Business Domain" *Book: The Essentials of Machine Learning in Finance and Accounting, Routledge, Taylor & Francis*, 2021 [**PDF**]

[C1] **Md Shajalal** and Masaki Aono "Health Information Retrieval" *Book: Signal Processing Techniques for Computational Health Informatics, pp(193-207), Springer-Verleg, part of Springer Nature* 2020 [**PDF**]

Scholarships

Marie Sklodowska Curie Fellowship Nov 2021 – Oct 2024
Awarded by European Union for PhD research in eXplainable AI under GECKO Project, a H2020 Marie Sklodowska Curie Action (MSCA) ITN at Fraunhofer FIT, Germany

MEXT (Monbukagakusho) Scholarship Oct 2015 – Sep 2017
Awarded by MEXT (Ministry of Education, Culture, Sports, Science and Technology) for International Masters Program in Computer Science & Engineering at Toyohashi University of Technology, Japan.

University Merit Scholarship Academic year: 2009/10 – 2010/11
Awarded for academic excellence during B.Sc.(Engg.) studies at University of Chittagong, Bangladesh

Awards

Jul 2026 | **Best Paper Award**
Best paper in AI in HCI at **28th HCI International 2026**, Montreal, Canada
Prize money of **2500 Euro with Certificate** [Link
Sponsored by **Industrie- und Handelskammer Siegen**

Dec 2025 | **Best PhD Dissertation Award**
The University of Siegen, Germany
Prize money of **2500 Euro with Certificate** [News & Photo]
Sponsored by **Industrie- und Handelskammer Siegen**

Dec 2016 | **Best Presentation Award**
12th Asia Information Retrieval Societies Conference 2016 (**AIRS2016**), China

2013 | **Divisional Champion**
ACM-ICPC Dhaka Site Regional Programming Contest held at NSU, Dhaka, 2013

2012 | **First Runner Up**
First Inter University Project Show Case at **Dhaka University IT Society** (DUITS)

2011 | Honorable Mention
ACM-ICPC Dhaka Site Regional Programming Contest held at NSU, Dhaka, 2011

Research Training

[TS1] **GECKO Winter School 2021,** hosted by University of Strathclyde, Glasgow, UK
[TS2] **GECKO Summer School 2022**, hosted by Fraunhofer FIT, Germany
[TS3] **GECKO Winter School 2023,** hosted by Aalborg University, Copenhagen
[TS4] **GECKO Summer School 2023** hosted by KTH Royal Institute of Technology
[TS5] **GECKO Winter School 2024** hosted by National Technical University of Athens, Greece

Reviewer / PC Member

PC Member
1. The 3rd International Conference on XAI (xAI2025), Istanbul, Turkey
2. The 2nd International Conference on XAI (xAI2024), Malta
3. The 39th ACM/SIGAPP Symposium On Applied Computing **(SAC2024)**, Spain
4. International Behavioral and Social Computing 2023, Larnaca, Cyprus
5. Intl. Conference on speech and Language Technologies (SPELLL) 2022, 2023
6. IEEE ICAICT 2018, Dhaka, Bangladesh

Reviewer

Journal

1. IEEE Transaction on Fuzzy Systems, [IEEE]
2. IEEE Transcation of Artificial Intelligence [IEEE]
3. Progress in AI [Springer]
4. Concurrency and Computation: Practice and Experience [Wiley]
5. Expert Systems [Wiley]
6. IEEE Access [IEEE]
7. The Journal of Supercomputing [Springer]
8. Asia-Pacific Journal of Operational Research [World Scientific]
9. Technological Forecasting and Social Changes [Elsevier]
10. Machine learning with applications [Elsevier]
11. Decision Support Systems [Elsevier]
12. Knowledge and Information Systems, [Springer]

Conference

1. Springer xAI2025, Istanbul, Turkey
2. ACM SAC 2024, Avila, Spain
3. HICSS 2024, Hawai, USA
4. ECML-PKDD 2023, Turin, Italy
5. International Behavioral and Social Computing, [IEEE]
6. IEEE ECCE Conference [IEEE]
7. SPELL 2022, 2023 [IEEE]
8. IEEE ICCIT 2018
9. IEEE ICAICT 2018

Conference Talks

April 2026 | **Invited talk at IT Security Minds 2025**, Koln, Germany

Jul 2026 | **The 28th HCI International 2026**, Montreal, Canada

Jul 2025 | **The 3rd World Conference on Explainable Artificial Intelligence (xAI2025)**
Istanbul, Turkey

Sep 2025 | **Mensch und Computer 2025**
Chemnitz, Germany

Jul 2024 | **Springer 2024**
The 2nd World Conference on Explainable Artificial Intelligence (xAI2024)
Valetta, Malta

Oct 2023 | **IEEE DSAA2023**
The 10th IEEE International Conference on Data Science and Advanced Analytics (DSAA)
Thessaloniki, Greece

July 2023 | **Springer xAI 2023, Lisbon, Portugal**
The 1st World Conference on eXplainable Artificial Intelligence (xAI 2023)
Lisbon, Portugal

Dec 2022 | **Springer SPELLL-2022, Tamil Nadu, India**
International conference on Speech & Language Technology for Low-resource Languages
Tamil Nadu, India

Nov 2022 | **ACM BuldSys-2022, Boston, MA, USA**
The 9th ACM International Conference on Systems for Energy-Efficient Built Environments
Boston, MA, USA

Jul 2022 | **ACM e-Energy-2022, Boston, MA, USA**
The 13th ACM International Conference on Future Energy Systems
Boston, MA, USA [**Slide**]

Feb 2020	**Springer ICONCS-2020, Dhaka, Bangladesh** International Conference on Cyber Security and Computer Science Dhaka, Bangladesh [**Slide**]
Sep 2018	**IEEE ICECE-2018, Dhaka, Bangladesh** International Conference on Electrical and Coumputer Engineering Dhaka, Bangladesh [**Slide**]
Sep 2018	**IEEE ICBSLP-2018, Sylhet, Bangladesh** International Conference on Bangla Speech and Language Processing Sylhet, Bangladesh [**Slide**]
Aug 2016	**IEEE ICAICTA-2016, Penang, Malaysia** International Conference on Advanced Informatics: Concepts, Theory and Application, Penang, Malaysia. [**Slide**]
Jul 2016	**The Twelfth NTCIR Conference** National Institute of Informatics (NII), Tokyo, Japan [**Slide**], [**Slide**]

Language

English	Fluent (Speaking, Reading and Writing)
German	**B2 (Upper Intermediate)**
Bengali	Mother tongue
Japanese	Basic Knowledge (Speaking and Listening)